Recommendations of 8 November: what was done and what came out

Z + jets in Summer22 (2022 C–D), same files as the 237-page deck

8 November 2026

Overview

#	recommendation	status	outcome	page
1	Muon discrepancy (trigger, η, SFs, Rochester)	done	Trigger SF fixed: μμ 0.966 → 0.943, ee 0.949	3–5
			pT shape: unchanged, same in ee (not muons)	
2	Other b/c taggers, L/M/T	done	DeepJet, RobustParT added; similar purity	6
3	Loose tau ID veto	postponed	–	8
4	Why ee and μμ pileup differ	done	Test files cover different runs	7
5	ee vs μμ MC colours (legend)	done	One stack and legend order for both	8
6	MET up to ~200 GeV	done	MC excess 40–120 GeV (known)	8
7	Zoom into high-statistics bins	done	10 zoomed plots in every region	8

Everything compares directly with the 237-page deck.

Same files and lumisections. All 237 Z→ee histograms are bit-for-bit identical, and the old muon setup reproduces the old Z→μμ yields exactly (data 65259). Statistical errors only, 20 files per sample.

1. Muons: what was checked

Each line is a separate muon setup in the same run, changing one thing. Right: how much data/MC would change with that choice instead of the new default.

suspect	other choice	after all cuts	before jet/MET cuts
Trigger SF	was: leading muon only (old choice)	+2.75%	+2.78%
	other POG method: leading muon matched to trigger	-0.81%	-0.51%
Trigger turn-on	second muon pT > 25 (old) instead of 26	-0.29%	-0.12%
	leading muon pT > 26 instead of 35	-0.01%	-0.37%
η cut	|η| < 2.1 instead of 2.4	+0.01%	+0.25%
Rochester	no muon scale/smearing (Run 3 replacement)	-0.24%	-0.15%
SFs	no ID + isolation SFs	-4.04%	-4.11%
	no pileup weights	+1.51%	+0.39%

One real problem: the trigger SF. IsoMu24 fires if either muon fires, so the SF must use both muons.

Now the POG two-muon formula with the data and MC efficiencies of muon_Z.json, and both muons pT > 26 GeV (where the SFs start).
The trigger-matching method agrees within 1%. |η| < 2.4 and the scale/smearing were already in.

1. Muons: outcome

data/MC per selection step, after the change.

step	Z→ee	Z→μμ
all Z events (no jet or MET cut)	0.995 ± 0.005	0.986 ± 0.003
MET < 50, no jet	1.042 ± 0.006	1.030 ± 0.004
MET > 50, ≥ 1 jet	0.692 ± 0.027	0.679 ± 0.019
final: MET < 50, ≥ 1 jet	0.949 ± 0.012	0.943 ± 0.009

Z→μμ data/MC 0.966 → 0.943. It now matches Z→ee (0.949) at every step.

The remaining 5–6% deficit after the cuts is common to both channels: MC has too many events with jets and far too many with MET > 50 GeV.

Open: muons with |η| > 2.0 are low before the jet/MET cuts (0.964 ± 0.010 vs 0.998 in the barrel).

Both POG trigger SF methods see it (trigger matching: 0.940). Cutting at |η| < 2.1 changes the total by only +0.25%.

1. Muons: the pT(ℓℓ) slope

data/MC per pT(ℓℓ) range divided by the overall data/MC (shape only), full selection.

pT(ℓℓ) [GeV]	0–30	30–60	60–100	100–150	150–200	200–300	300–400	400–1000
Z→μμ, 237-page deck	1.020	0.975	1.015	1.041	0.996	0.982	0.805	1.011
Z→μμ, today	1.017	0.978	1.012	1.039	0.996	0.984	0.805	1.016
Z→ee (unchanged)	1.032	0.968	1.012	1.042	1.033	0.969	0.898	0.639
events, Z→μμ today	16808	25207	13323	5068	1498	777	123	61

The visual slope is still there, unchanged. No muon setup changes it, and Z→ee has the same pattern.

data/MC peaks at 100–150 GeV and falls above 200 GeV. Before the jet and MET cuts both channels already fall at high pT(ℓℓ) (Z→ee -2.1 ± 1.2, Z→μμ -3.5 ± 0.8 % per 100 GeV): DY modelling, not muons.

The fitted slope of +4.4% per 100 GeV came from the run with the DY Z pT weights on.

Same fit, same files: weights on +4.4 ± 1.3; 237-page deck (weights off) -0.7 ± 1.2; today -0.8 ± 1.2. A straight line misses the up-and-down pattern, so the fit is near zero.

2. Other taggers: DeepJet and RobustParT next to PNet

Leading jet, POG working points, each tagger with its own BTV SFs. Purity = share of real b (c) jets in the tagged DY MC (Z→μμ). Tag ratio = tagged fraction in data / in MC, Z→ee and Z→μμ.

tagger	WP	b-tag purity	b-tag ratio ee / μμ	c-tag purity	c-tag ratio ee / μμ
PNet	L	22%	0.998 / 0.980	9%	0.981 / 0.985
	M	62%	1.011 / 0.969	20%	0.989 / 0.985
	T	88%	1.058 / 0.925	51%	0.965 / 0.963
DeepJet	L	20%	0.976 / 0.979	9%	1.015 / 1.016
	M	58%	1.056 / 0.996	18%	0.994 / 1.025
	T	87%	1.050 / 0.929	48%	1.088 / 1.014
RobustParT	L	23%	1.013 / 1.040	9%	1.014 / 1.019
	M	62%	1.079 / 1.006	20%	1.021 / 1.045
	T	89%	1.034 / 0.963	54%	1.099 / 1.051

Similar purity at the same WP. b-tag: data and MC agree for all three (largest 1.7 σ).

c-tag: average tag ratio PNet 0.978, DeepJet 1.025, RobustParT 1.042 (largest 2.4 σ), so the taggers differ by a few percent there.
Stat. errors 1–8%; no SF uncertainties yet.

4. Why the ee and μμ pileup plots differ

Pileup of the lumisections our 20 data files cover (brilcalc, normtag, 69.2 mb), vs the full 2022 C–D profile the MC is weighted to.

	mean pileup, files	full era	data − MC, mean N(PV)	expected
Z→ee (EGamma files)	32.8	33.8	-0.57	-0.64
Z→μμ (Muon files)	33.6	33.8	-0.14	-0.14

Expected = pileup difference × N(PV) per pileup interaction from the MC (0.69).

The test files cover different runs, and that predicts each channel's N(PV) offset.

The EGamma files have less high pileup than the Muon files and the full era.

Caveat: reweighting the MC bin by bin to each subset over-corrected, so it is not used.

The clean test is the full-dataset run, where both channels see the same runs.

3, 5, 6, 7: the other items

5. ee vs μμ MC colours: fixed.

The plot script sorted the stack by each channel's own yields, so DY+b and DY+c swapped places in Z→ee. Now one fixed order for both channels. The 237-page deck's plots were remade too.

6. MET up to 200 GeV: done (new plot without the MET cut, in every region).

MC is well above data from 40 to 120 GeV in both channels (data/MC down to about 0.6): the 2022 jet/MET issue traced in October. Not fixed.

7. Zoom: done.

Zoomed copies (linear scale, ratio 0.8–1.2) of Z pT, both lepton pTs, leading and second jet pT, generator Z pT, MET, N(PV), dijet mass and pT balance, in every region.

3. Loose tau ID veto: postponed, as agreed.

Still open, to decide, and where everything is

Still open

• Common 5–6% deficit after the jet and MET cuts (MC excess with jets and with MET > 50 GeV).
• pT($\ell\ell$) shape (dip at 30–60 GeV, fall above 200 GeV) in both channels.
• Muons with |η| > 2.0 about 3% low. • Tau veto. • SF uncertainties. • Only 20 files per sample.

To decide

Keep the new muon setup (two-muon trigger SF, pT > 26) as the default? Which tagger and working point? Then run the full Summer22 dataset.

Files (in hists_zbzc_run3_v2/Summer22/)

slides_zbzc_v2_Summer22.pdf: answers deck with plots (25 pages).
slides_zbzc_btagL_3taggers_Summer22.pdf: Loose b-tag, three taggers, key plots (65 pages).
plots_<region>/: all 49 plots for each of the 19 regions. Tables: muvar/, pu/, tagger_summary.txt, channels_summary.txt.